

HD189733b base breathing: localization of the residual and why base-BC / IC-blend fixes do not help

EXHALE relaxes HD209458b-like hot Jupiters to a steady wind, but HD189733b does not converge: the time-marching settles into a bounded limit cycle (the “base breathing”), and the steady (Newton/PTC) solvers stall — even when started from a converged, spectrum-matched Wind-AE warm-start initial condition. This memo records a per-cell localization of the steady residual, three targeted fixes that all turned out to be null, and the resulting conclusion: the obstruction is not the base boundary condition nor the initial-condition blend, but an intrinsic mismatch between the Wind-AE solution and EXHALE’s discretized momentum equation, amplified by the residual normalization. It is recorded so the dead ends are not re-explored and the live options are clear.

1. Symptom

With a Wind-AE warm-start (IC mode: windae, Solver: Newton) HD189733b shows *no* NaN but never converges. The steady residual $\|R\|$ descends through a violent transient (peaks ~ 400) to ~ 0.15 , then erupts back to ~ 2.4 and repeats — a sustained limit cycle that never reaches the Newton-finish threshold (5×10^{-2}), so the JFNK/PTC hand-off never fires (or, when forced, stalls). HD209458b and other less strongly bound planets converge normally with the identical machinery.

2. Diagnostic: a per-cell residual dump

residual_norms (steady_residual.f90) reports only the per-component *maximum* relative residual; it hides *where* the imbalance lives. A per-cell dump was added to the ATES_RESIDUAL=1 path (EXHALE_main.f90), writing output/residual_profile.txt with columns $r, n, v, T, R_{\text{mass}}, R_{\text{mom}}, R_{\text{energy}}$ for every cell $j = 1 \dots N$. This was the decisive tool.

3. Localization

The steady residual of the loaded (banked, spectrum-matched) HD189733b Wind-AE IC is *purely momentum*: the reported component norms are $R_{\text{mass}} = 2.2 \times 10^{-3}$, $R_{\text{mom}} = 4.63$, $R_{\text{energy}} = 5.2 \times 10^{-2}$. The per-cell profile (Table 1) shows two distinct facts:

1. **The base peak is excluded.** R_{mom} is largest at the very base ($r \approx 1.0002$, ~ 180 , set by the gravity source ρg at the dense anchor $n_0 = 10^{14}$) and decays outward. But residual_norms takes $\max |R(j_{\min} : N)|$ (steady_residual.f90 L58), and $j_{\min}$ sits above the base cluster. *The base cells the breathing “lives” on are not in the metric the solver minimizes.*
2. **The metric max is the Wind-AE region.** The reported $\|R_{\text{mom}}\| = 4.63$ comes from $r \approx 1.05$, which is *above* the windsoln base $R_{\min} \approx 1.030$ — i.e. in the interpolated Wind-AE solution, not the base blend. There the absolute residual is only ~ 0.088 , but the wind momentum $\rho v \approx 0.019$ is tiny, and the relative norm $0.088/0.019 \approx 4.6$.

Table 1: Per-cell momentum residual of the HD189733b Wind-AE IC. The windsoln base radius is $R_{\min} \approx 1.030 R_{\text{pl}}$; the convergence metric uses only $[j_{\min} : N]$ (here $j_{\min}$ is at $r \approx 1.05$).

$r [R_{\text{pl}}]$	R_{mom}	in the $[j_{\min} : N]$ metric?
1.0002	1.8×10^2	no (base, $j < j_{\min}$)
1.0041	1.3×10^2	no
1.0079	9.5×10^1	no
1.0501	8.8×10^{-2}	yes ($\approx j_{\min}$) — the metric max
1.10	8.6×10^{-3}	yes

4. Root cause

The metric residual is therefore the *interpolated Wind-AE solution failing to satisfy EXHALE’s own discretized momentum equation* at the wind base ($r \sim 1.05$, the densest, slowest part of the wind, where the normalization is most punishing). The two codes genuinely differ — mean molecular weight / EOS, the gravity discretization, the HLLC flux and reconstruction, the 2-D Rate/2 + Mdot/2 averaging. The Wind-AE IC is a valid Wind-AE solution but is *not* EXHALE’s steady state, and for the strongly bound HD189733b the gap is large enough (after normalization) that neither time-marching (limit cycle) nor the implicit solvers can close it. It is *not* a property of the base ghost cells nor of the sub- $R_{\min}$ blend.

5. Fixes attempted — all null

Each fix below left the metric byte-identical at $k=2 = 4.630339$, because each one acted somewhere other than $r \approx 1.05$ (Table 2).

6. What was added (gated, working cases byte-identical)

- **Per-cell residual dump** — the `ATES_RESIDUAL=1` path writes `output/residual_profile.txt`. The diagnostic that localized everything above; kept.
- **Hydrostatic base: flag** (`hydrostatic_base`, default off $\Rightarrow$ byte-identical legacy base BC). When on, the ghost pressure is the momentum-consistent extrapolation of (A). Null for HD189733b but harmless and physically reasonable for coarser grids; kept off.
- **Roche IC blend** (`wae_ic_writer.f90`: `phiR` + the `wae_write_ic` `Mrapp, atilde, tidalf` arguments). With `tidalforce=0` it is identical to the legacy spherical blend, so spherical runs (e.g. the HD209458b example) are unaffected; for tidal runs it makes the sub- $R_{\min}$ base layer Roche-consistent. Not the breathing lever, but a correct improvement; kept.

Table 2: Targeted fixes for the HD189733b base breathing and why each was null. “ k_2 ” is the reported momentum residual norm.

Attempt	What it changed	Why it was null
(A) momentum-consistent ghost p	ghost pressure = linear extrapolation of the interior dp/dr (vs legacy fixed $p \rightarrow T_0$), so dp/dr is continuous at the base	base cells are excluded from $[j_{\min} : N]$; metric unchanged.
(A') isothermal-hydrostatic ghost	ghost $\rho, p = \rho_{\text{bc}} e^{-(\phi(r) - \phi_{\text{base}})/c_s^2}$ (Roche ϕ), so $dp/dr = -\rho g$ exactly	ghosts sit at $r \approx 1.0000$ (adjacent to the base), so the factor ≈ 1 ; and the base is excluded anyway.
(③) Roche IC blend	the IC-writer sub- $R_{\min}$ density blend uses the Roche potential ϕ_R (scaled by tidalforce) instead of the spherical $b_{0,\text{eff}}(1/r - 1)$	the metric residual is <i>above</i> $R_{\min} \approx 1.030$; the blend only touches $r < R_{\min}$.
PTC / JFNK from the IC	direct pseudo-transient / Jacobian-free Newton steady solve (ATES_PTC=1)	stalls at $\ R\ \approx 4.6$, $d\tau$ frozen, line-search $\lambda \rightarrow 10^{-7}$ (no descent direction).
+ frozen base (ATES_PTC_NFIX)	anchor the first few base cells	same stall.
+ smooth valve (Valve_eps)	differentiable base velocity valve	same stall.

7. Conclusion and live options

The HD189733b base breathing is *not* curable by the base boundary condition or the IC blend: the obstruction is the Wind-AE $\leftrightarrow$ EXHALE discretization mismatch at the wind base, amplified by the small- ρv normalization of the $[j_{\min} : N]$ residual. The remaining paths, all larger than a localized patch, are:

1. **Solver / normalization.** A more robust steady solver (better globalization than the current PTC/JFNK, which lose the descent direction here), and/or a residual normalization that does not blow up where the wind momentum ρv is tiny.
2. **Discretization match.** Bring EXHALE’s base momentum discretization (EOS, gravity, flux) closer to Wind-AE’s so a Wind-AE solution is genuinely near EXHALE’s steady state.
3. **Accept known-hard.** Treat HD189733b (strongly bound) as a case EXHALE does not currently relax, while HD209458b-like planets converge.